

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.36322838	delta=+0.0116308	
ratio=1.0049459			
CoG x [m]	-0.002024708562 -> -0.0338177741	delta=-0.0317931	ratio=-
CoG y [m]	-3.052657894e-05 -> -0.03721546035	delta=-0.0371849	
ratio=-			
CoG z [m]	0.009509749599 -> 0.01164150702	delta=+0.00213176	
ratio=-			
rotor effectiveness 1	1 -> 1.062631985	delta=+0.062632	ratio=1.062632
rotor effectiveness 2	1 -> 0.9710517395	delta=-0.0289483	ratio=0.97105174
rotor effectiveness 3	1 -> 0.8506376677	delta=-0.149362	ratio=0.85063767
rotor effectiveness 4	1 -> 0.9322686783	delta=-0.0677313	ratio=0.93226868
rotor command lag [s]	0.004957675934 -> 0.05106355197	delta=+0.0461059	
ratio=-			

Inertia matrix [kg m^2]

reference:

[0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[1.901508003e-05, 5.9167305e-08, 0.1289921107]

estimated:

[0.06599522231, -0.0008591827513, -0.0008539327834]
[-0.0008591827513, 0.0655553494, 0.0003766106364]
[-0.0008539327834, 0.0003766106364, 0.1289743454]

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06488759602	delta=-6.50491e-05
ratio=0.99899851		
principal moment 2 [kg m^2]	0.06500006701 -> 0.06664902293	delta=+0.00164896
ratio=1.0253685		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1289882982	delta=-3.81815e-06
ratio=0.9999704		

Optimization

joint dynamics loss: 569.5820307
Gaussian-prior cost: 13.660132328
selected strict-ZOH rotor lag [s]: 0.051063551966

Bag failure_2_single

reconstruction position RMSE [m]: 47.3227094668
reconstruction position component RMSE [m]: [9.003482 5.058639 46.182099]
reconstruction orientation RMSE [deg]: 141.383508606
gyro consistency RMSE xyz [rad/s]: [17.435623 17.072428 23.259355]
specific-force consistency RMSE xyz [m/s^2]: [51.236391 61.762413 33.63222]
inferred external force RMS xyz [N]: [7.121741 14.323964 5.150213]
inferred external torque RMS xyz [N m]: [2.864997 3.58412 8.271362]
bag dynamics loss: 569.5820307
gimbal command lag [s] 0.004957199097 -> 0.004181744776

Recorded command timestamp intervals (data-derived)

Bag failure_2_single
rotor: median dt=0.00495767593 s (201.707 Hz), min=2.14576721e-06 s, max=0.15852499 s
gimbal: median dt=0.0049571991 s (201.727 Hz), min=2.62260437e-06 s, max=0.158331871 s

Selected optimizer diagnostics
status=0 success=False nfev=80 njev=6
message=The maximum number of function evaluations is exceeded.
reported optimality=7.88752095104428e-06
initial cost=583.2421632430004
final cost=583.2421630275993
cost reduction=2.1540108718909323e-07
coordinate step L2=0.00028055998027753763
final gradient inf-norm=7.88752095104428e-06
gradient tolerance satisfied=False
final Jacobian rank=14
final Jacobian cond(nonzero)=324.730746272064
max directional finite-difference Jacobian relative error=5.1508985782697236e-12